ICT2113: Software Modelling & Analysis
Project Specification
[2025/26 T2]

Project Team P2 - 0602C
Blockchain Pharmaceutical Supply Chain Verification System

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1. Company Description

Medicare Plus is a Singapore based pharmaceutical company engaged in the manufacturing, wholesale distribution, and retail supply of prescription and over-the-counter medicines. Over the past decade, Medicare Plus has experienced rapid growth in both product volume and geographic reach. While this expansion has increased market presence, it has also introduced significant complexity into the company's supply chain operations.

Currently, Medicare Plus manages its supply chain with paper-based documentation while the suppliers, manufacturers, wholesalers, distributors, transporters and retail outlets maintain their own records.This means that the data integrity is inconsistent across the different organisations and the tracking cycle of medicine from production to final customer might lead to delays and potential orders getting missed out or giving wrongly.

Furthermore, Medicare Plus has experienced counterfeit products and unauthorised diversion of medicines and inventory discrepancies, which damages the company's reputation and can lead to regulatory penalties, accompanied with challenges such as proving the authenticity and origin of medicines during audits. There is also the concern of slow operations such as the order creation process due to the need for manual intervention and increased administrative overhead.

To address these challenges, Medicare Plus is seeking a secure, transparent, and automated digital solution that enables end-to-end traceability of pharmaceutical products. The company aims to adopt a system that leverages transparent digital technology to automatically enforce business rules, record transactions immutably, and ensure that every stakeholder in the supply chain operates on a single trusted source of truth.

Mission

To provide safe, high-quality, and affordable pharmaceutical products while strengthening supply chain integrity through the adoption of secure, automated, and transparent information systems that protect patient safety and enhance operational efficiency.

Vision

To become the most trusted pharmaceutical manufacturer and distributor by ensuring medicinal authenticity, traceability managed through transparent digital technologies.

Core Business

- Procurement of pharmaceutical raw materials from certified suppliers
- Manufacturing of generic and branded medicines in compliance with regulatory standards
- Wholesale distribution of medicines to hospitals, clinics, and pharmacies
- Retail supply of pharmaceutical products through authorized distributors
- Coordination of logistics and transportation services
- Quality control, compliance reporting, and product traceability

Key Initiatives Related to New Software

- End-to-End digital traceability that records every transaction and movement of medicines from manufacturing to customer end.
- Business process automation for order creation, verification, approval, from manufacturers to customers.
- Anti-counterfeiting and source verification of the origin of medicines by accessing tamper proof transaction histories.
- Process digitisation and paperless operations to reduce errors, delays and administrative costs.
- Compliance and audit readiness by maintaining immutable records to simplify regulatory audits and improve compliance reporting.

2. Project Description

2.1.Project Name

The project will be “Smart Contract Enabled Pharmaceutical Supply Chain System for Anti-Counterfeiting and Product Traceability”

2.2 Context

Pharmaceutical supply chains are increasingly complex, involving multiple independent stakeholders such as suppliers, manufacturers, wholesalers, distributors, transporters, and healthcare providers, which makes it difficult to maintain consistent security, transparency, and accountability across the medicine lifecycle.

Moreover, most stakeholders operate separate information systems and still rely on paper-based documentation, resulting in fragmented data, limited traceability, and time-consuming manual reconciliation when incidents occur. The weak verification mechanisms and unauthorised distributors further enable counterfeit and falsified medicines to enter legitimate supply channels, posing serious risks to patient safety and undermining public trust. At the same time, traditional centralised databases are vulnerable to tampering, insider threats, and cyberattacks, making it difficult to guarantee data integrity.

These challenges are compounded by inefficient manual processes for order placement, verification, approvals, and status updates, which can delay the availability of critical medicines. In addition, increasing regulatory requirements for end-to-end traceability, secure record keeping, and timely reporting place further strain on organisations using legacy systems. Collectively, these issues highlight the need for a secure, transparent, and tamper-resistant system that provides end-to-end visibility, automates supply chain processes, and enables reliable verification of pharmaceutical products.

2.3 Project Goals (SMART)

Specific

This project aims to design and implement a blockchain based pharmaceutical supply chain system that will provide end-to-end traceability of medicines from suppliers to consumers. The system will utilise smart contracts to automate order creation, verification, approvals, and status updates among the suppliers, manufacturers, wholesalers, distributors, transporters and customers. Additionally, it will also provide a trusted system for verifying the authenticity and source of pharmaceutical products, while ensuring secure role based access to system functionalities to maintain data integrity.

Measurable

In order to ensure that the progress does not drift from its requirements, there will be measurable goals set in place to ensure the project is delivered fully. This goals will be split according to its respective main requirements starting with at least 95% of the supply chain transactions are recorded on the blockchain to provide single source of truth, manual processing steps for orders and verification are reduced by at least 60%, and 90% of unauthorised or unverified product entries are detected and prevented during testing, furthermore, 100% of the users will be able to retrieve medicine traceability information, lastly, there needs to be at least 80% positive feedback achieved during user acceptance testing before the release of the final product.

Attainable

The project will be attainable by using the appropriate and reliable resources with a budget of $1,500,000. This will be achievable by using existing and proven technologies such as recognised blockchain platforms, smart contracts, cryptographic hashing and a decentralised database. The system involves HTML, CSS and JavaScript in the front end and ASP.net with C# and the IDE used is Visual Studio, which is widely used and is cost efficient. The smart contract used will be “Stratis Smart Contract” which is free for development cost and only requires cost when being deployed which is another aspect in lowering the cost. The scope of the project is limited to core supply chain processes, making implementation feasible within the available time and resources.

Relevant
The project directly addresses critical challenges in pharmaceutical supply chains, including counterfeit medicines, lack of traceability, data tampering, and operational inefficiencies. By improving security, transparency, and automation, the system supports patient safety, regulatory compliance, and organisational efficiency, making it highly relevant to current industry needs.

Time-Bound
The project will be completed within a period of 13 months, with major milestones starting with Requirements Approval & Design Completion taking 2 months, while the Blockchain Infrastructure & Core Modules development will take 3 months, and Integration & User Interface Completion takes 3 months, the UAT Completion & Security Audit Approval will take 2 months, and finally the release of production will take 2 months paired with the final documentation transfer which will take up 1 month.

3. Requirements Tree

REQ ID	REQUIREMENT DESCRIPTION
RT1.0	**BLOCKCHAIN INFRASTRUCTURE**
RT1.1	Stratis blockchain platform integration with Cirrus Core sidechain
RT1.1.1	SHA-256/SHA-512 cryptographic hash functions for block integrity verification
RT1.1.2	Proof-of-Authority consensus mechanism for transaction validation
RT1.2	Smart contract deployment and management system
RT2.0	**USER ROLE MANAGEMENT SYSTEM**
RT2.1	Administrative Module: Account approval, smart contract assignment, tamper detection
RT2.2	Supplier Module: Order management, raw material tracking, transporter assignment
RT2.3	Manufacturer Module: Medicine production, raw material procurement, order verification
RT2.4	Wholesaler Module: Order processing, distributor fulfillment, transporter coordination
RT2.5	Distributor Module: Wholesaler order management, customer order processing
RT2.6	Customer Module: Medicine search/ordering, source verification, order tracking
R3.0	**AUTHENTICATION AND SECURITY**
RT3.1	Multi-factor authentication for all accounts with role-based access control
RT3.2	Public/private key cryptography for digital signatures on blockchain transactions
RT4.0	**DATA MANAGEMENT AND STORAGE**
RT4.1	Microsoft SQL Server for off-chain data: user profiles, product catalogs
RT4.2	Blockchain ledger for transaction hashes, custody records and audit trails
RT5.0	**USER INTERFACE AND EXPERIENCE**
RT5.1	Responsive web interface using google chrome / explorer
RT5.2	Laptop or PC with windows 7 or higher

4. Business & Operational Requirements

4.1 Assumptions (A)

ID	Assumption	Notes / Implication
A-01	MediCare Plus will operate a permissioned blockchain network	Limits participants to vetted stakeholders
A-02	All participants are known, registered, and vetted entities	Supports trust model/onboarding controls
A-03	Blockchain is used for verification and audit, not physical enforcement	Still needs physical anti-counterfeit controls
A-04	Transporters are third-party logistics providers	Requires transporter onboarding + tracking
A-05	Users have basic computer literacy and can complete training	Training plan needed
A-06	Compatible with any browser	Implies cross-browser testing scope

4.2 Company Constraints (C) & Rules (R)

ID	Statement	How to verify
C-01	Must operate within cybersecurity policies, including AES-256 encryption	Security audit / encryption-at-rest + in-transit evidence
C-02	Integration must not disrupt current operations during deployment	Cutover plan + rollback + staging/UAT signoff
C-03	Max downtime: 4 hours/month during off-peak (2–6 AM EST)	Monitoring + SLA report (also align timezone)
R-01	All medication orders must be verified by authorized personnel before shipment	Workflow enforcement + role-based approvals
R-02	Dual authorization for high-value shipments exceeding **$50,000**	Threshold rule + 2-person approval logs (also clarify currency)

4.3 External Business Systems

System Name	Purpose	Integration Method
Epic EHR System	Patient prescription validation	HL7/FHIR API
SAP ERP	Inventory management, financials	REST API with OAuth 2.0
Omnicell Automated Dispensing	Medication dispensing verification	Vendor API connector
FDA Sentinel System	Adverse event reporting	HTTPS data submission

4.4 Primary Business Systems

- Pharmaceutical procurement
- Distribution logistics
- Inventory tracking
- Compliance auditing

4.5 Government & International Regulations

- Ministry Of Health (MOH)
  - Healthcare Services Act 2020 Confidentiality of patient/medical information - licensing conditions for healthcare institutions
  - Cyber & Data Security Guidelines (2023) Cybersecurity controls for healthcare IT systems

- Health Science Authority (HSA)
  - Good Distribution Practice (GDP) Full traceability required - documentation of batch numbers, expiry dates, quantities, supplier/purchaser details
  - Health Products Act 2007 All therapeutic products must be registered - manufacturers, importers, wholesalers must hold valid licenses.
  - Good Manufacturing Practice (GMP) compliance for all manufacturers - GMP certification verification required

- Infocomm Media Development Authority (IMDA)
  - Electronic Transactions Act Legal validity of smart contracts and electronic signatures

- Personal Data Protection Commission (PDPC)
  - Personal Data Protection Act 2012 Consent, purpose limitation, data protection, breach notification, 7-year retention

- ISO 9001 (Quality Management)
- ISO 27001 (Information Security)

4.6 Business Requirements (BR)

ID	Name of Req.	Stakeholder	Constraints & Business Rules	Requirement Clause	Priority
BR-01	User Registration & Account Creation	Direct End Users (Customers, Healthcare Providers)	Users must register with a unique email. Passwords must be in secure formatting (min 12 chars, complexity requirements). Email verification required	Users must create an account to access system functionalities Account approval managed by Administrator role	**Essential**
BR-02	Login Authentication	Direct End Users	Only registered users and admins can log in. Password recovery must be secure via email verification with time-limited tokens.	Provide login authentication with 'Forgot Password' functionality via email recovery Multi-factor authentication (MFA) required for administrative accounts	**Essential**
BR-03	Blockchain Infrastructure Management	System Administrators	Must operate a blockchain network. Only vetted stakeholders can participate. Minimum 5 validator nodes required for network resilience	Admins manage Stratis blockchain platform integration with Cirrus Core sidechain for smart contract execution. Monitor blockchain integrity and detect tampering attempts.	**Essential**

BR-04	Smart Contract Deployment	System Administrators	Smart contracts created manually in Core Dashboard initially. Contracts Immutable once deployed.	Admins create and assign smart contracts to user accounts. Contracts written in C# and deployed to Stratis platform for automated verification and custody transfer management.	**Essential**
BR-05	Supplier Order Management	Suppliers	**Role-based access:** Only authorized suppliers can manage raw material orders. Must track batch numbers and expiry dates.	Suppliers view orders from manufacturers, manage raw materials inventory, assign transporters for shipments, and update order status with blockchain verification.	**Acceptable**
BR-06	Manufacturer Medicine Production	Manufacturers	Must comply with Good Manufacturing Practice (GMP) certification. All products require serialization with New Distribution Capability (NDC) codes.	Manufacturers manage medicine production, add raw materials, create/update/delete orders for raw materials, verify orders, and initiate supply chain with blockchain transactions.	**Desirable**
BR-07	Wholesaler Distribution	Wholesalers	Licensed wholesalers only. Must maintain Good Distribution Practice (GDP) compliance. Dual authorization for shipments >$50,000 SGD.	Wholesalers manage orders from manufacturers, process distributor orders, verify order authenticity, update delivery status, and assign transporters with blockchain custody records.	**Acceptable**
BR-08	Distributor Order Fulfillment	Distributors	Must verify all orders against blockchain records. Temperature controlled shipments require IoT sensor integration for supply chain monitoring.	Distributors manage orders from wholesalers, process customer orders, verify product authenticity via blockchain, update status, and coordinate last delivery with transporters.	**Acceptable**
BR-09	Customer Medicine Ordering	End Customers (Healthcare Providers, Patients)	Customers must be registered and verified. Prescription medications require valid prescription verification through Epic EHR integration.	Customers search for medicines, place orders, verify source and authenticity via blockchain QR code scanning, view order history, and track shipment status in real-time.	**Acceptable**

BR-10	Medication Source Verification	End Customers, Healthcare Providers	Verification must check the blockchain for complete custody chain. QR codes must use cryptographic signatures to prevent counterfeiting.	Customers and providers verify medication source by scanning QR codes to view complete supply chain history from manufacturer to delivery, stored immutably on blockchain.	**Essential**
BR-11	Transporter Logistics Management	Third-Party Logistics Providers	Transporters must be registered, or vetted entities. GPS tracking required for all shipments. Real-time updates are mandatory.	Transporters receive assignments from suppliers/wholesalers/distributors, update location and delivery status, confirm custody transfers with digital signatures on blockchain.	**Desirable**
BR-12	Role-Based Access Control (RBAC)	All User Roles	Principle of least privilege enforced. Users can only access functions relevant to their role. Admin approval required for role changes.	Enforces role-based permissions for Admin, Supplier, Manufacturer, Wholesaler, Distributor, Customer, and Transporter modules with segregation of duties.	**Essential**
BR-13	Custody Transfer Recording	All Supply Chain Participants	Every custody transfer must be recorded on blockchain with digital signatures from both parties. Timestamp must be in UTC format.	Records all custody transfers (supplier→manufacturer, manufacturer→wholesaler, wholesaler→distributor, distributor→customer) as immutable blockchain transactions.	**Essential**
BR-14	Order Status Tracking	All User Roles	Status updates must be real-time (within 5 seconds of blockchain confirmation). Automated notifications via webhooks for status changes.	Users can view and update order status: Pending, Approved, In Production, Shipped, In Transit, Delivered, Verified Status changes trigger blockchain transactions and notifications	**Desirable**
BR-15	Product Master Data Management	System Administrators, Manufacturers	NDC codes must be validated against the HSA database. Product information must include batch number, expiry date, quantity, and manufacturing location.	Admins and manufacturers manage pharmaceutical product catalog with NDC codes, descriptions, specifications, serialization data, and regulatory information.	**Desirable**

BR-16	Regulatory Compliance Reporting	Compliance Officers, Administrators	Must generate reports compliant with MOH Healthcare Services Act, HSA Good Distribution Practice, and PDPC Personal Data Protection Act requirements.	Generates audit trails, compliance reports for regulatory submissions, adverse event reports, and transaction history with 7-year retention for HSA and PDPC compliance.	**Essential**
BR-17	Integration with Epic EHR	Healthcare Providers, Administrators	Integration must use HL7/FHIR R4 standards. Patient data must be encrypted and comply with confidentiality requirements per Healthcare Services Act 2020.	Integrates with Epic EHR for prescription validation, medication administration records, and patient safety checks using secure HL7/FHIR APIs.	**Desirable**
BR-18	Integration with SAP ERP	Finance Teams, Administrators	Integration must maintain transaction integrity. Financial data must be reconciled daily. OAuth 2.0 authentication required.	Integrates with inventory management, financial transactions, procurement workflows, and automated invoice reconciliation using REST APIs.	**Desirable**
BR-19	Tamper Detection and Alerts	System Administrators	Real-time monitoring of blockchain integrity. Alerts must be sent within 30 seconds of detecting suspicious activity. Alert must include transaction hash and involved parties.	Admins receive real-time alerts if any blockchain block or transaction has been tampered with or manipulated. The system automatically locks affected accounts pending investigation.	**Essential**
BR-20	Batch and Lot Number Tracking	Manufacturers , Distributors, Regulatory Compliance	Must track all batch/lot numbers per HSA Good Distribution Practice. Documentation must include supplier, purchaser, quantity, and dates.	The system tracks batch numbers, lot numbers, expiry dates, and quantities for full traceability. Supports recalls by batch/lot number with one-click notification to all affected parties.	**Essential**

5. Technical Requirements (TR)

ID	Name of Reqr.	Remarks
5.1 Hosting Solution		
TR-01	Cloud Infrastructure	Secure cloud infrastructure hosted on Microsoft Azure with 99.9% uptime SLA, geographic redundancy across multiple data centers, and automated disaster recovery capabilities

TR-02	Production Environment	Production environment hosted in US East region with backup replication to US West region for business continuity
TR-03	Development/Staging	Separate development, staging, and UAT environments with production-like configuration for testing and validation
5.2 Server Architecture		
TR-04	Three-Tier Architecture	Presentation layer (web UI), Application layer (business logic and APIs), Data layer (databases and blockchain nodes) with clear separation of concerns
TR-05	Load Balancing	Load-balanced application servers with auto-scaling capabilities to handle traffic spikes (minimum 2 servers, scale up to 10)
TR-06	Blockchain Node Servers	Dedicated servers for blockchain validator nodes with minimum specifications: 16 GB RAM, 1 TB SSD, 8-core CPU
5.3 Choice of Platform or CMS		
TR-07	Application Framework	ASP.NET Core 6.0 framework with MVC (Model-View-Controller) pattern for web application development
TR-08	No Traditional CMS	Custom-built application without traditional content management system, content managed through admin portal
TR-09	Development IDE	Microsoft Visual Studio 2019 Professional Edition or higher with ASP.NET and web development workloads
5.4 Administration Tools		
TR-10	Admin Portal	Web-based administration portal for system configuration, user management, smart contract deployment, and monitoring dashboards with role-based access
TR-11	Command-Line Interface	CLI tools for advanced administrative operations, blockchain node management, and automated deployment scripts
TR-12	Monitoring & Logging	Centralized logging with Azure Application Insights or equivalent, real-time monitoring dashboards for system health, transaction volume, and blockchain status
5.5 Online Payment Providers		
TR-13	Payment Processing	Handled through existing SAP ERP integration, direct payment gateways not required for Phase 1
TR-14	Future Payment Integration	Future phases may integrate payment gateways (Stripe, PayPal, or regional providers) for direct consumer purchases if B2C functionality is added
5.6 Integration Constraints		
TR-15	HL7/FHIR Compliance	Must support HL7/FHIR R4 standards for healthcare interoperability with Epic EHR system
TR-16	API Response Time	Maximum API response time: 2 seconds for 95th percentile requests, 5 seconds for 99th percentile

TR-17	Integration Testing	Integration testing required with Epic EHR test environment before production deployment
TR-18	Backward Compatibility	APIs must maintain backward compatibility for minimum 12 months after version deprecation
5.7 Computer Language		
TR-19	Backend Language	C# 10.0 with .NET 6.0 for server-side application logic and business rules
TR-20	Frontend Languages	JavaScript ES6+, HTML5, CSS3 for client-side user interface and interactions
TR-21	Smart Contract Language	C# for Stratis blockchain smart contracts compiled to .NET bytecode
TR-22	Database Query Language	T-SQL for Microsoft SQL Server queries, stored procedures, and database operations
5.8 Data Security Management		
TR-23	Encryption at Rest	AES-256 encryption for all data stored in databases and file systems
TR-24	Encryption in Transit	TLS 1.3 for all network communications including API calls, web traffic, and blockchain synchronization
TR-25	Access Controls	Role-based access control (RBAC) with principle of least privilege, multi-factor authentication (MFA) for all administrative accounts
TR-26	Audit Logging	Comprehensive audit logging for all data access, modifications, and administrative actions with 7-year retention
TR-27	Backup & Recovery	Automated daily backups with point-in-time recovery capability, backup retention: 30 days online, 7 years archived
5.9 Maintenance		
TR-28	Scheduled Maintenance Windows	Maximum 4 hours per month during off-peak hours (2:00-6:00 AM EST) for planned maintenance
TR-29	Automated Deployment	CI/CD pipeline using Azure DevOps or GitHub Actions for automated testing and deployment
TR-30	Blue-Green Deployment	Blue-green deployment strategy for zero-downtime upgrades where possible, rollback capability within 15 minutes
TR-31	Patch Management	Security patches applied within 48 hours of release, feature updates deployed monthly during maintenance windows
5.10 Migration		
TR-32	Data Migration Tools	ETL (Extract, Transform, Load) tools for importing existing medication catalog, user accounts, and historical transaction records from legacy systems
TR-33	Legacy System Integration	Ongoing synchronization with legacy systems during 6-month transition period using message queues and scheduled batch jobs
TR-34	Migration Validation	Data validation and reconciliation processes to ensure 100% accuracy of migrated records, automated comparison reports

TR-35	Rollback Plan	Documented rollback procedures to revert to legacy systems if critical issues encountered during migration
5.11 Browser or Operating System Compatibility		
TR-36	Supported Browsers	Chrome 100+, Firefox 100+, Microsoft Edge 100+, Safari 15+ with automatic detection and graceful degradation for older versions
TR-37	Operating System Compatibility	Compatible with Windows 10/11, macOS 12+, and modern Linux distributions for desktop access
TR-38	Mobile Responsive Design	Mobile-responsive web design supporting iOS 15+ and Android 11+ on smartphones and tablets (1024x768 minimum resolution)
TR-39	Cross-Browser Testing	Automated cross-browser testing as part of CI/CD pipeline, manual testing on all supported browsers before production release
5.12 Expected Performance (Loading Times, General Speed)		
TR-40	Initial Page Load	Initial page load time: <3 seconds for desktop, <5 seconds for mobile on 4G connection
TR-41	Subsequent Interactions	Subsequent page interactions: <1 second response time for standard operations (search, view, update)
TR-42	Concurrent Users	System must support minimum 500 concurrent users without performance degradation
TR-43	Daily Transaction Volume	Handle 2,500+ pharmaceutical transactions per day with peak capacity of 500 transactions per hour
TR-44	Blockchain Confirmation Time	Blockchain transaction confirmation: <30 seconds for standard transactions, <60 seconds for complex smart contract executions
TR-45	Database Query Performance	95% of database queries complete in <500ms, complex analytical queries <5 seconds

6. Financials

6.1 Payment Schedule (Percentage-Based)

For the payment schedule the prices are redacted and are displayed in percentages. The project will kick off once contract is signed and there is an upfront payment of 15% payment for this milestone, followed by the planning stage which will be 10% upon requirements approval and design completion, after that the development phase will be split into 2 payments of 25% and subsequently 20%, once that is completed then the testing phase payment will be 15% once the user acceptance rate is satisfied, this will lead to the deployment phase payment of 10% when the application has gone live, the last payment of 5% will be paid out when the project has been fully handed over.

Phase	Milestone/Deliverable	Percentage	Timing
Initiation	Contract Signing & Project Kickoff	15%	Upon signing
Planning	Requirements Approval & Design Completion	10%	Month 2
Development Phase 1	Blockchain Infrastructure & Core Modules	25%	Month 5
Development Phase 2	Integration & User Interface Completion	20%	Month 8
Testing	UAT Completion & Security Audit Approval	15%	Month 10

Deployment	Production Go-Live & Initial Stabilization	10%	Month 12
Closure	Final Acceptance & Documentation Transfer	5%	Month 13

6.2 Estimated Cost Breakdown

Based on industry benchmarks for enterprise blockchain and healthcare IT systems, the total estimated project cost is approximately **SGD [Redacted]**. The breakdown are justified as such: the largest portion is allocated to software development as it involves building the core web application, smart contracts and blockchain integration, while the second highest cost is allocated to the infrastructure and cloud services as its used to keep the application running for the long term, there are some smaller costs such as the integration with external system which is used for APIs. The smaller components such as security and compliance, quality assurance testing, project management and training are costs that are needed for administrative costs.

Cost Category	Estimated Cost (SGD)	Percentage
Infrastructure & Cloud Services	[redacted]	20%
Software Development (Core System & Smart Contracts)	[redacted]	40%
Integration with External Systems (APIs)	[redacted]	13%
Security & Compliance (Audits, Encryption, Regulatory Alignment)	[redacted]	10%
Testing & Quality Assurance (UAT, Performance, Security Testing)	[redacted]	7%
Project Management & Documentation	[redacted]	7%
Training, Deployment & Change Management	[redacted]	3%
Total Estimated Project Cost	[redacted]	**100%**

7. Scheduling

7.1 Milestones Deliverables

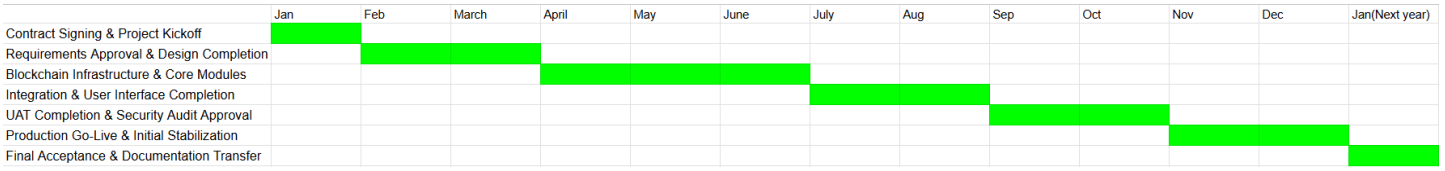

Fig. 1 Milestones Deliverables

7.2 Progress Reporting

- **Weekly Status Reports:**
  - Submitted every Friday by 5 PM SGT, documenting completed tasks, upcoming activities, risks, and issues

- **Monthly Steering Committee Meetings:**
  - First Tuesday of each month at 10 AM SGT, reviewing milestone progress, budget variance and change requests

- **Sprint Reviews:**
  - Bi-weekly demonstrations of completed functionality with business users and technical team members

- **Earned Value Reports:**
  - Monthly EVM analysis tracking Schedule Performance Index (SPI) and Cost Performance Index (CPI)

- **Risk Register Updates:**
  - Continuous monitoring with formal updates at each steering committee meeting

8. Sign-Off

8.1 Status

This specification is **validated and ready for approval**.

8.2 Authenticator

Project Manager - PharmaTrust Distributors Ltd.

Signature: Date: 30 Jan 2026

8.3 Designated Approvers (Not Yet Signed)

CIO - MediCare Plus Network

Signature: Date: 30 Jan 2026

Head of Compliance - MediCare Plus Network

Signature: Date: 30 Jan 2026

Technical Director - PharmaTrust Distributors Ltd.

Signature: Date: 30 Jan 2026

9. Addenda

9.1 Use Case Model (Graphical)

Actors:	Core Use Cases:
• Admin • Supplier • Manufacturer • Wholesaler • Distributor • Transporter • Customer	• Register Participant • Create & Verify Order • Assign Transporter • Transfer Custody • Verify Medicine Source • Audit Supply Chain

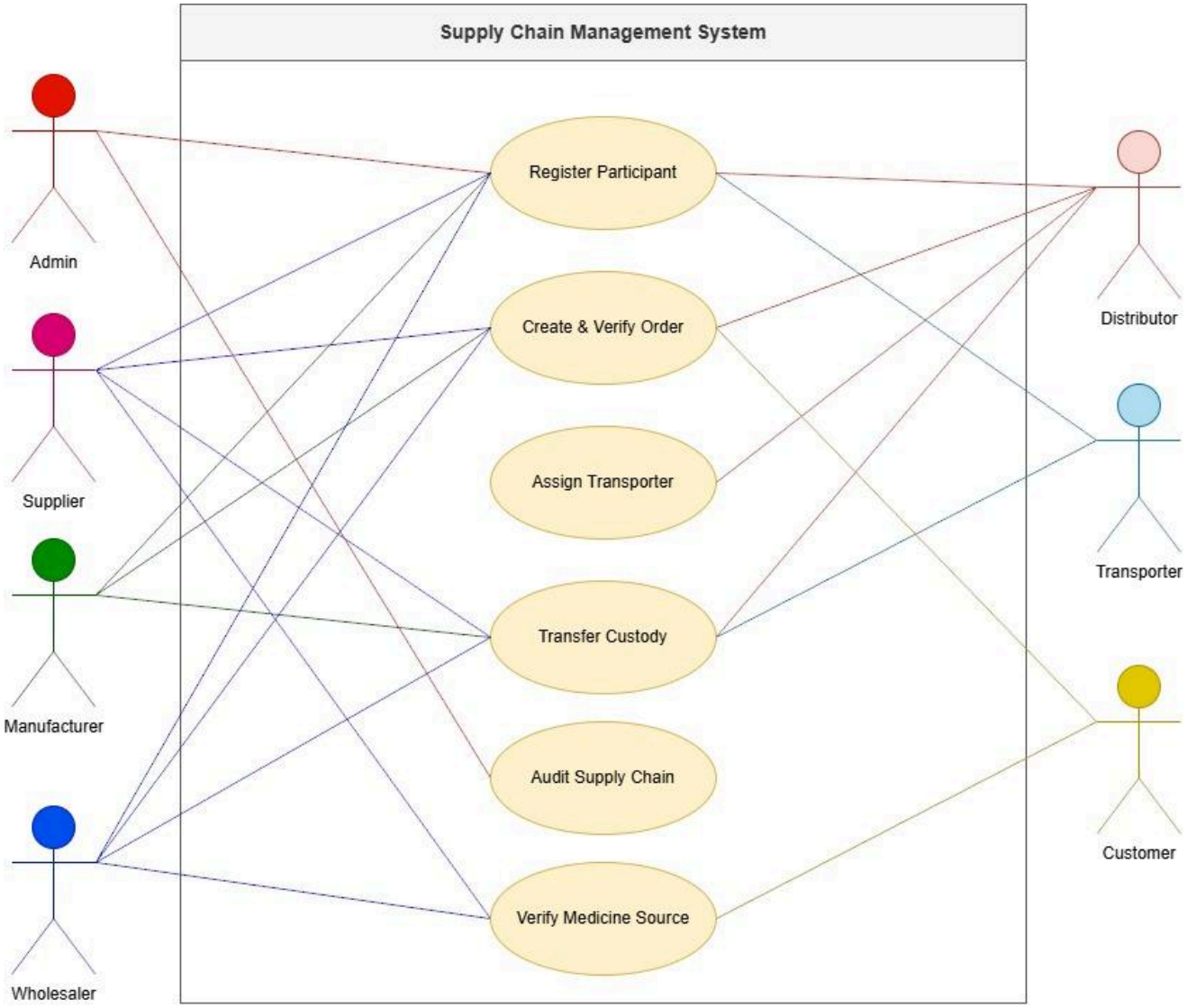

Fig. 2 Simple Use Case Diagram

9.2 Data Dictionary (DD)

The following table provides a representative subset of critical data elements.

Code	Element Name	Data Type	Length/Format	Description
DD-001	UserID	VARCHAR	50	Unique user identifier
DD-005	UserRole	ENUM	Fixed values	Admin, Supplier, Manufacturer, etc.
DD-012	NDC_Code	VARCHAR	11 (5-4-2)	National Drug Code
DD-018	SerialNumber	VARCHAR	20	DSCSA unit serial number
DD-023	LotNumber	VARCHAR	15	Manufacturing lot identifier
DD-045	TransactionHash	VARCHAR	64 (hex)	SHA-256 blockchain hash
DD-051	BlockNumber	BIGINT	8 bytes	Sequential block identifier
DD-067	SmartContractID	VARCHAR	40 (address)	Stratis contract address
DD-089	CustodyStatus	ENUM	Fixed values	Pending, In-Transit, Delivered
DD-112	TimestampUTC	DATETIME	ISO 8601	Transaction timestamp (UTC)

9.3 Organization Chart

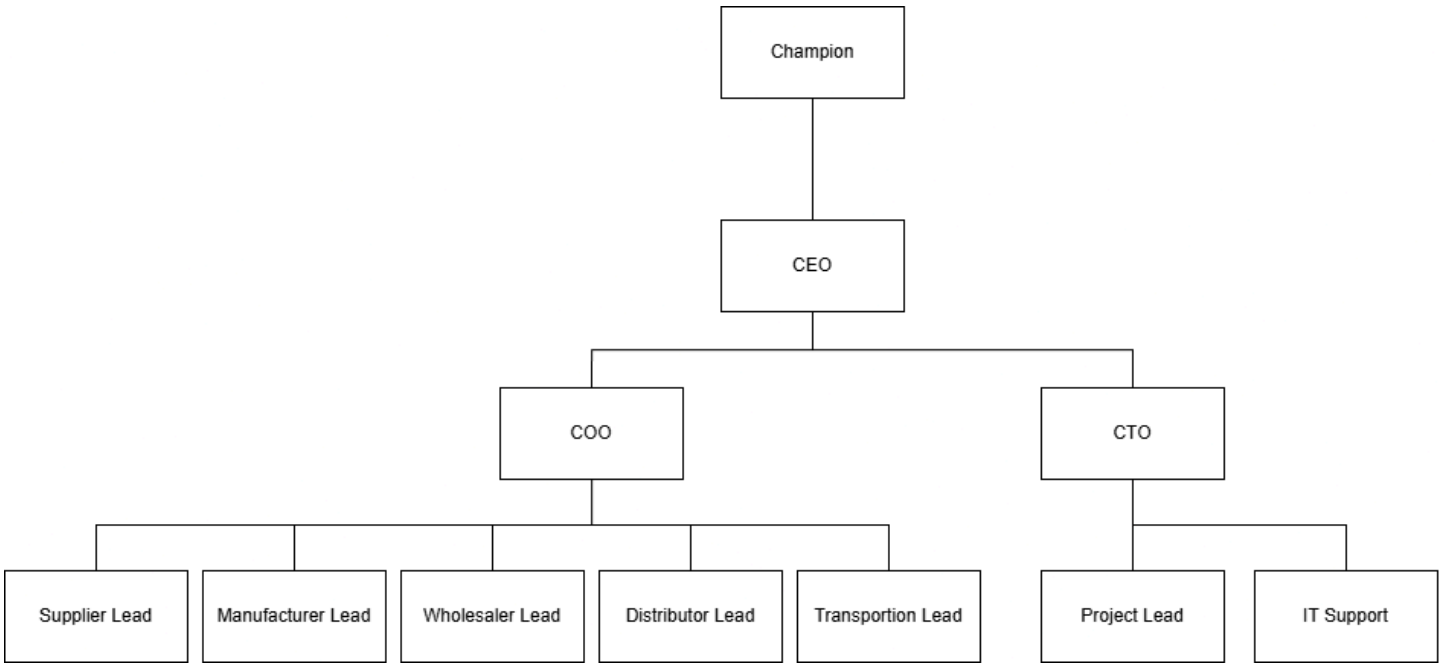

Fig. 3 Organization Chart

9.4 Wireframe

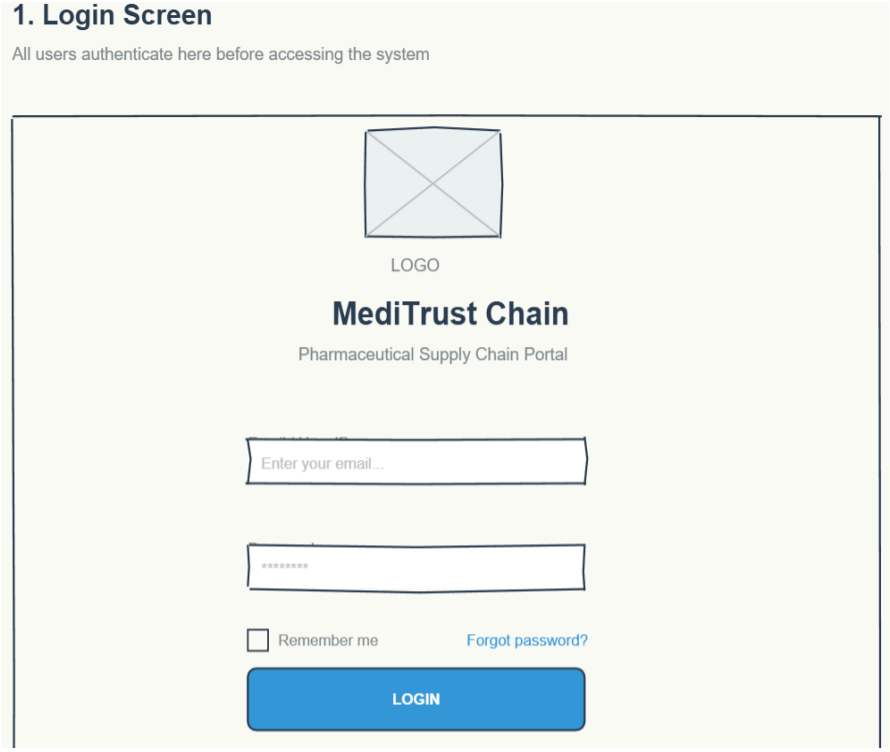

Fig. 4.1 Wireframe Login Screen

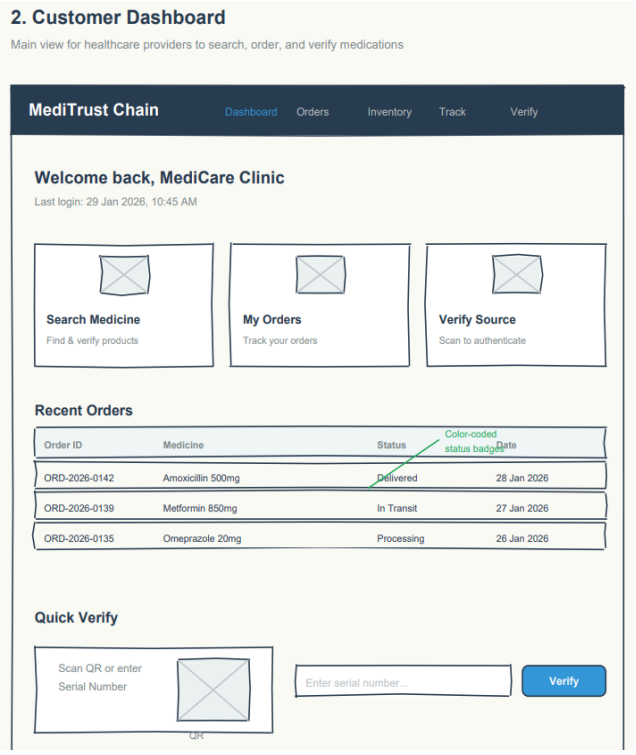

Fig. 4.2 Wireframe Customer Dashboard

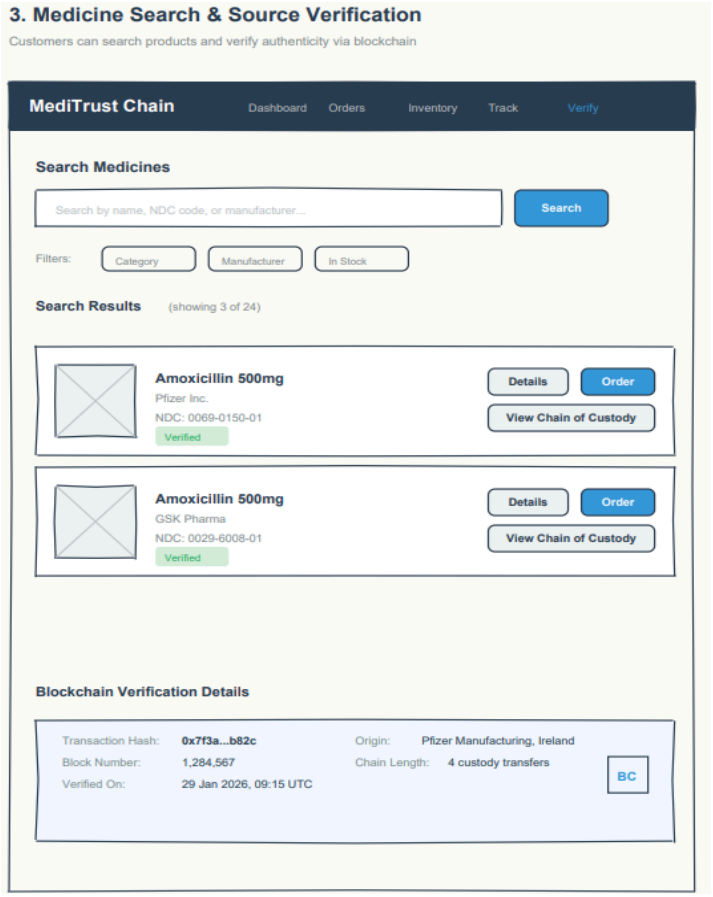

Fig. 4.3 Wireframe Medicine Search & Source Verification

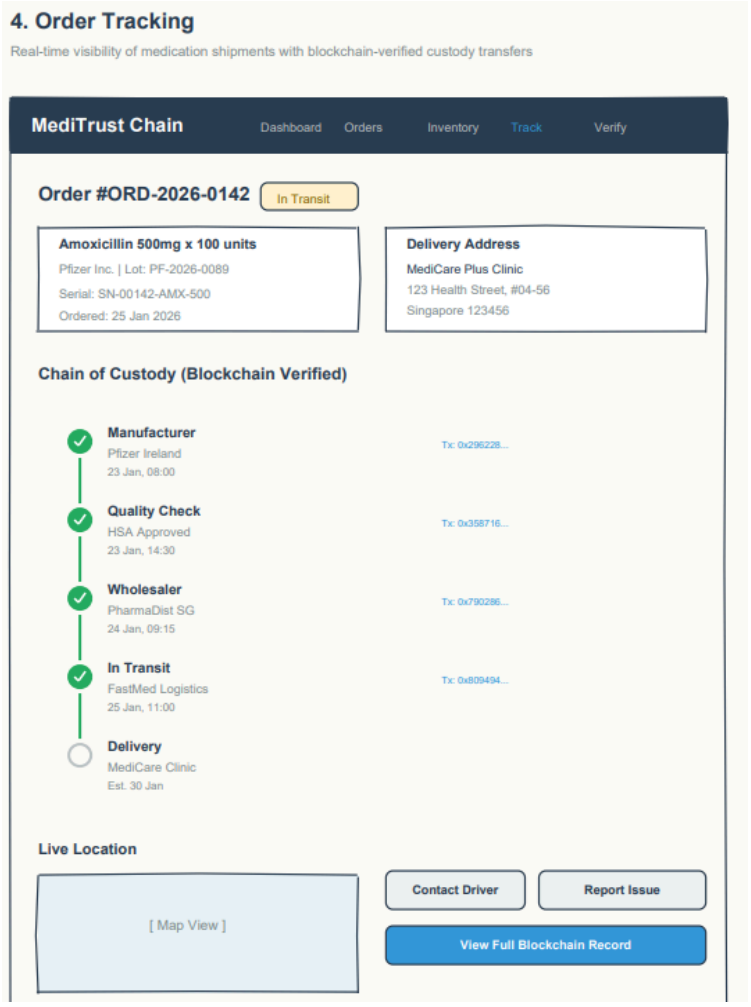

Fig. 4.4 Wireframe Order Tracking

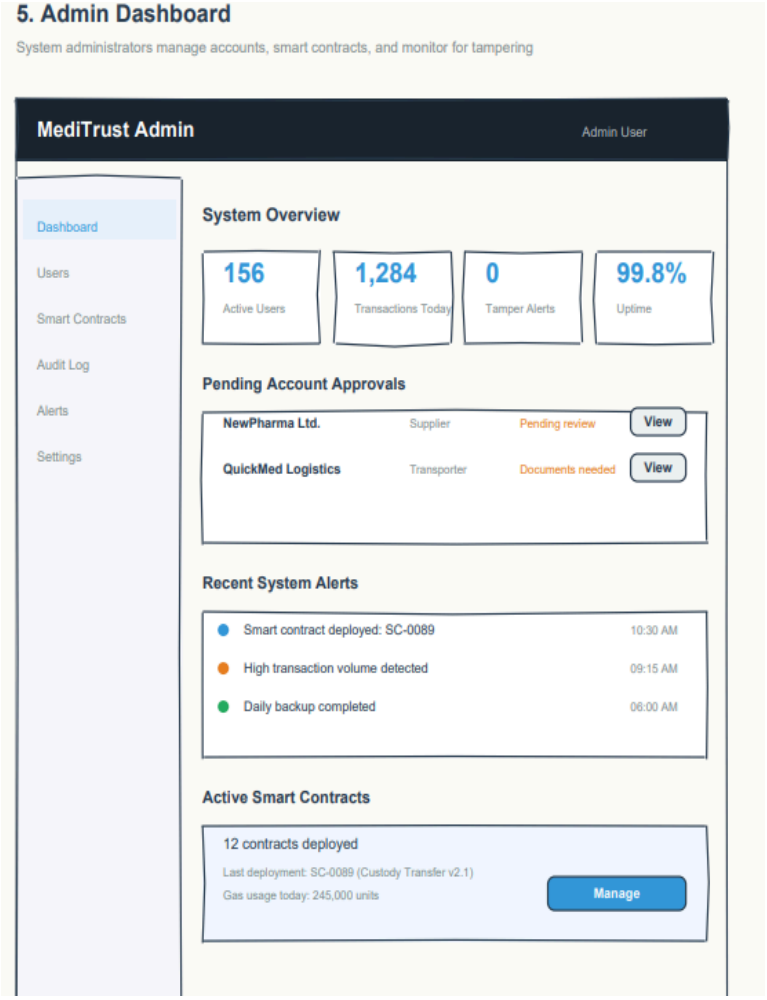

Fig. 4.5 Wireframe Admin Dashboard

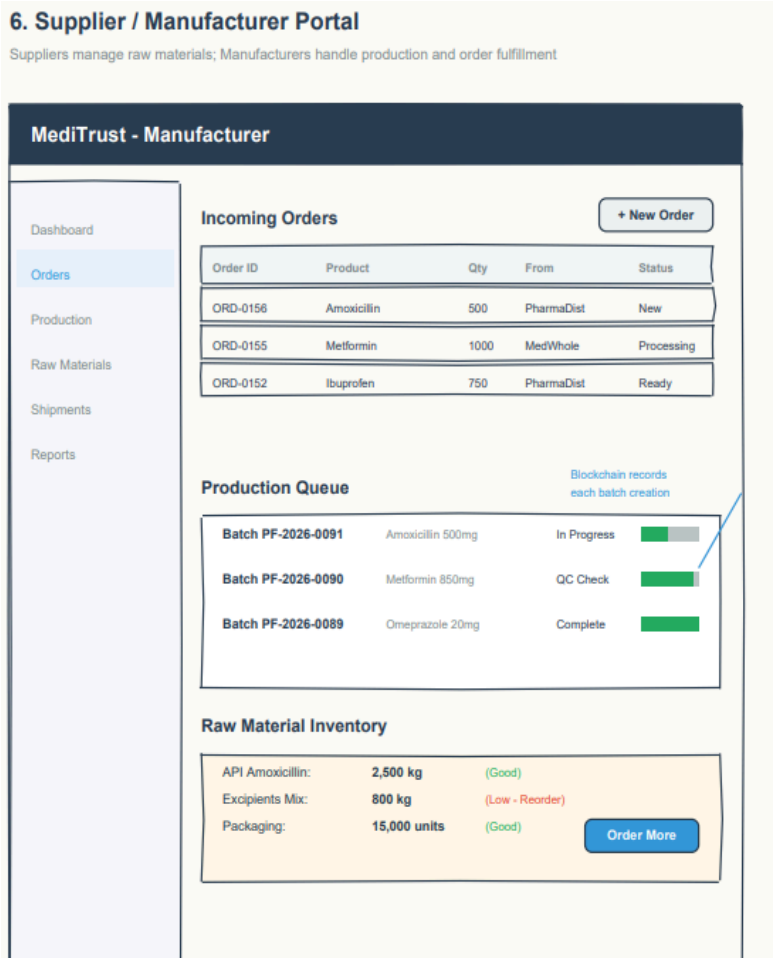

Fig. 4.6 Wireframe Supplier/Manufacturer Portal

9.5 Dialogue Map

9.5.1 Customer Flow

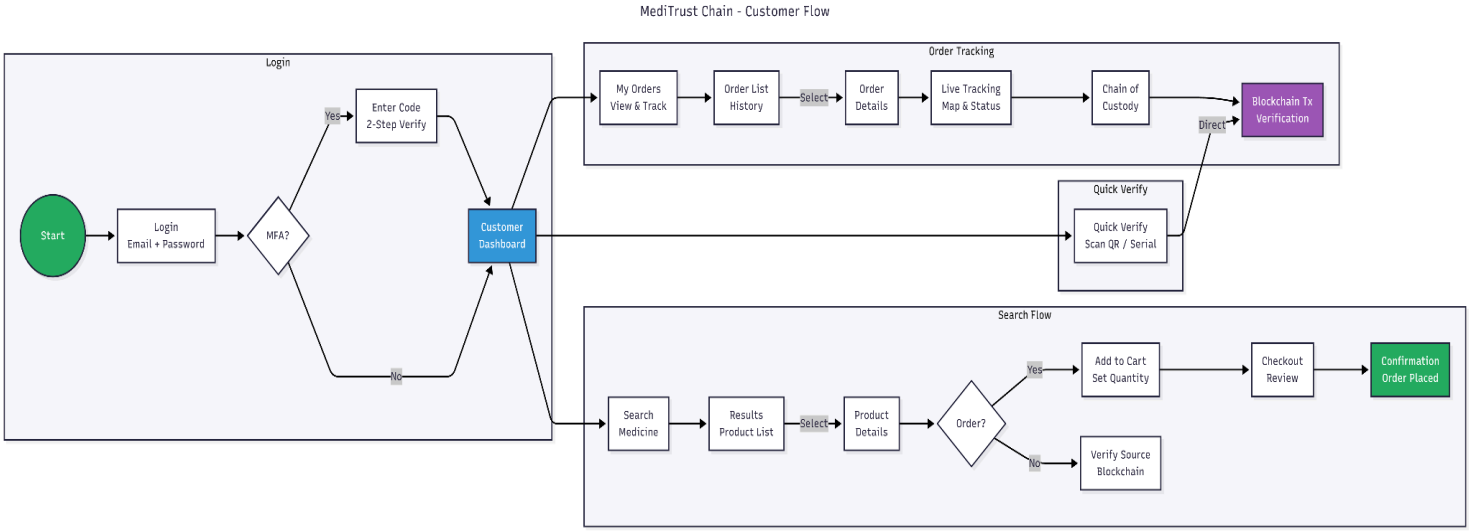

Fig. 5.1 Dialogue Map Customer Flow

9.5.2 Admin Flow

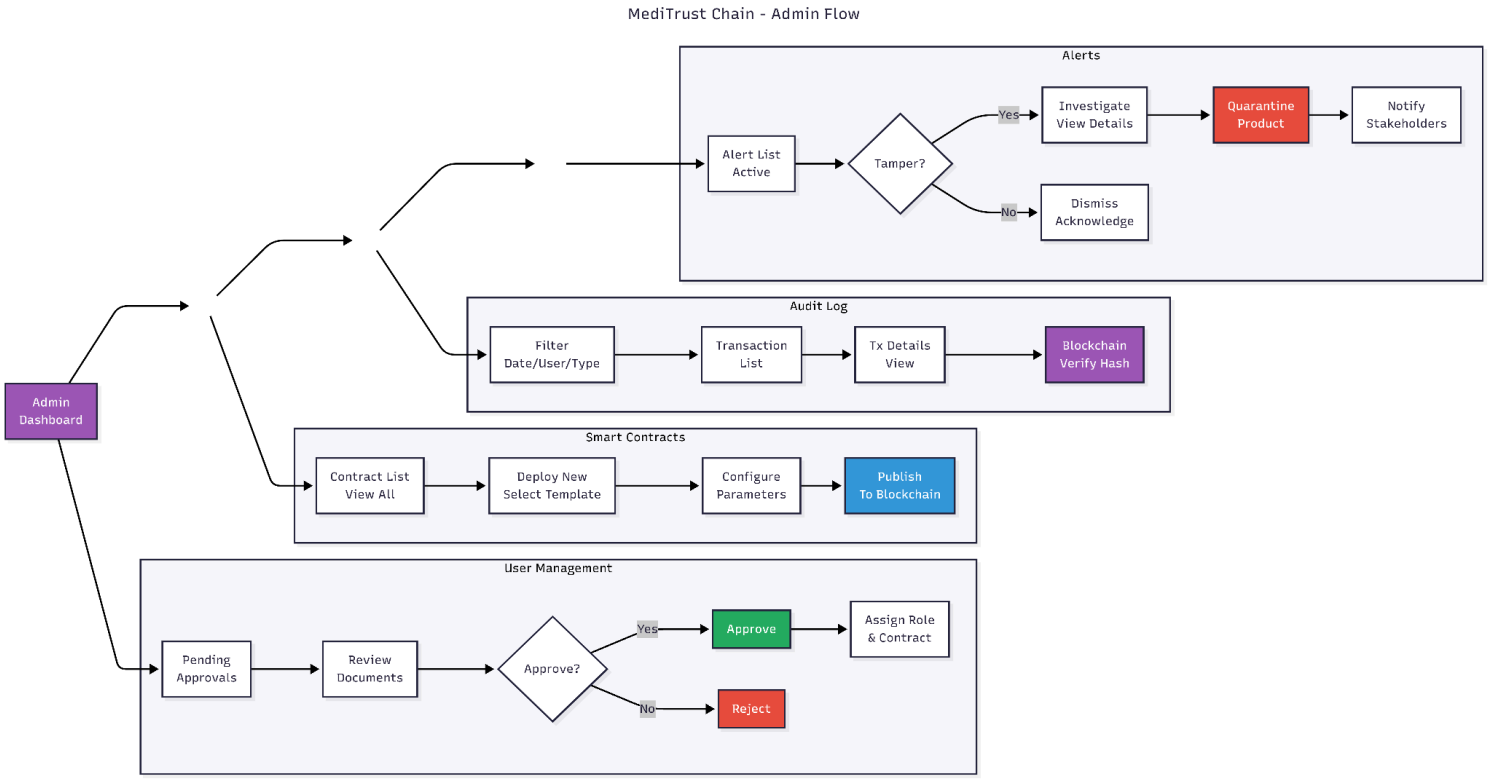

Fig. 5.2 Dialogue Map Admin Flow

9.5.3 Supplier Flow

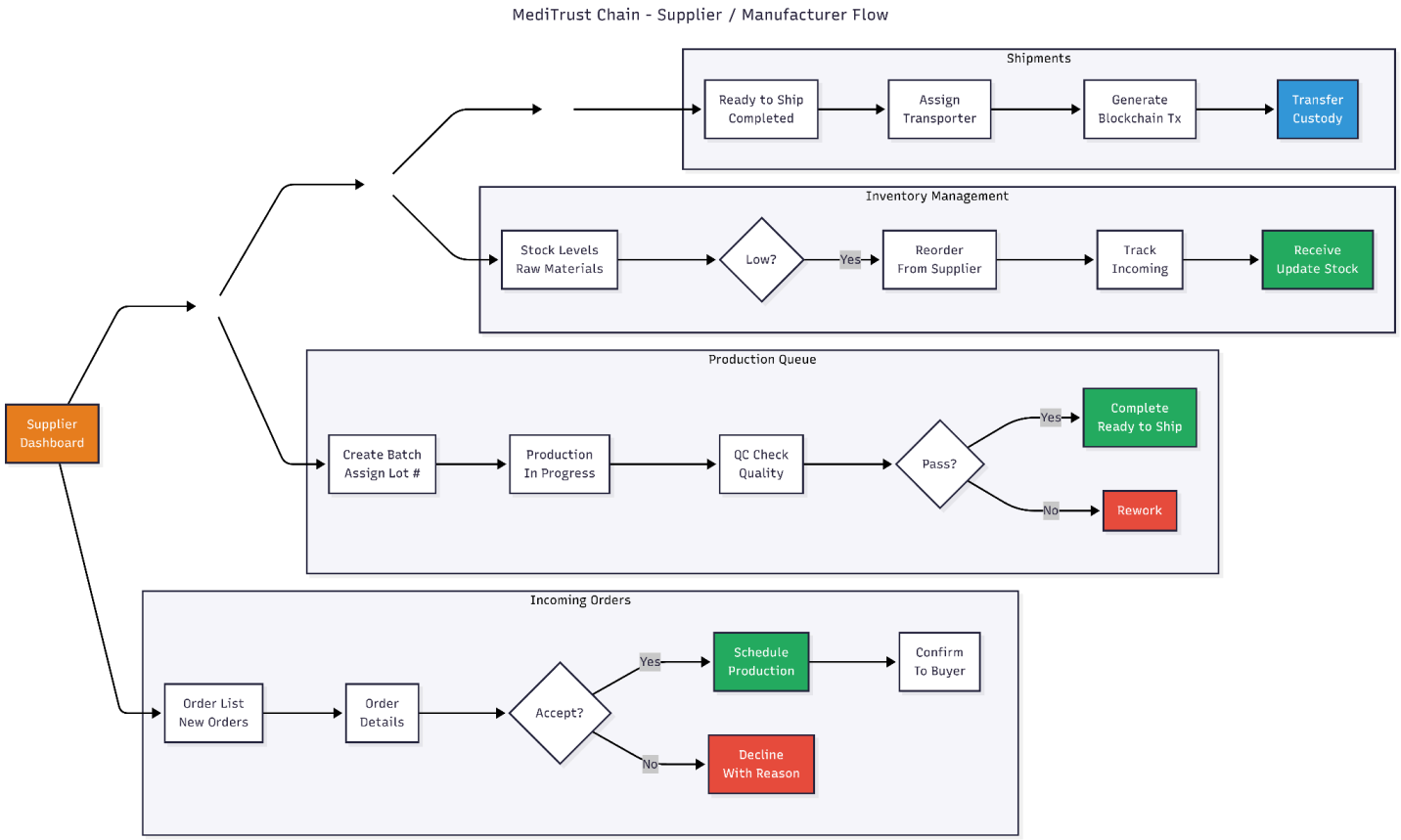

Fig. 5.3 Dialogue Map Supplier Flow

9.6 Key Clarifications

- Smart contracts are template-based and deployed programmatically, not manually
- Transporter is a recognized participant role, not a core admin module
- Blockchain stores hashes, MSSQL stores operational data
- System uses permissioned blockchain, not public chain
- Blockchain complements, not replaces, physical anti-counterfeit controls

9.7 Elicitation Plan

Technique	Target Stakeholders	Purpose	Justification
Interviews	Executives, IT leads, compliance officers	Understand strategic goals and constraints	Enables deep exploration of complex requirements
Workshops	Supply chain managers, technical teams	Map end-to-end workflows and pain points	Facilitates collaborative requirement discovery
Observation / Shadowing	Logistics teams, pharmacy staff	Understand real operational processes	Reveals implicit and undocumented requirements
Document Analysis	Policies, SOPs, regulatory documents	Identify compliance and procedural requirements	Ensures alignment with existing organisational standards
Surveys / Questionnaires	Healthcare providers and customers	Capture usability and user expectations	Enables scalable feedback from diverse users
Prototyping / Wireframes	End users and business stakeholders	Validate functional requirements and UI expectations	Reduces ambiguity and misinterpretation
Focus Groups	Selected stakeholders	Prioritise requirements and resolve conflicts	Encourages consensus-building

9.8 Key Stakeholders

Stakeholder Group	Role in Organisation	Type of Requirements
Executive Management	Strategic decision-making	Business goals, constraints
Compliance & Legal Team	Regulatory oversight	Regulatory & policy requirements
IT & Security Team	System design and security	Technical & non-functional requirements
Supply Chain Managers	Operational workflows	Process & operational requirements
Suppliers & Manufacturers	Upstream supply chain	Integration & data requirements
Distributors & Transporters	Logistics operations	Tracking & custody requirements
Healthcare Providers (Hospitals/Pharmacies)	Medication dispensing	Functional & usability requirements
Customers / Patients	End users	Verification & usability requirements
Government Agencies (HSA, MOH)	Regulatory authorities	Compliance requirements

END OF REPORT